

3.2.3 Waves

- **Progressive Waves**

Oscillation of the particles of the medium; amplitude, frequency, wavelength, speed, phase, path difference.

$$c = f\lambda$$

- **Longitudinal and transverse waves**

Characteristics and examples, including sound and electromagnetic waves.

Polarisation as evidence for the nature of transverse waves; applications e.g. Polaroid sunglasses, aerial alignment for transmitter and receiver.

- **Refraction at a plane surface**

Refractive index of a substance s , $n = \frac{c}{c_s}$

Candidates are not expected to recall methods for determining refractive indices.

Law of refraction for a boundary between two different substances of refractive indices n_1 and n_2 in the form

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Total internal reflection including calculations of the critical angle at a boundary between a substance of refractive index n_1 and a substance of lesser refractive index n_2 or air;

$$\sin \theta_c = \frac{n_2}{n_1}$$

Simple treatment of fibre optics including function of the cladding with lower refractive index around central core limited to step index only; application to communications.

- **Superposition of waves, stationary waves**

The formation of stationary waves by two waves of the same frequency travelling in opposite directions; the formula for fundamental frequency in terms of tension and mass per unit length is not required.

Simple graphical representation of stationary waves, nodes and antinodes on strings.

- **Interference**

The concept of path difference and coherence.

The laser as a source of coherent monochromatic light used to demonstrate interference and diffraction; comparison with non-laser light; awareness of safety issues.

Candidates will not be required to describe how a laser works.

Requirements of two source and single source double-slit systems for the production of fringes.

The appearance of the interference fringes produced by a double-slit system,

$$\text{fringe spacing } w = \frac{\lambda D}{s},$$

where s is the slit separation.

- **Diffraction**

Appearance of the diffraction pattern from a single slit.

The plane transmission diffraction grating at normal incidence; optical details of the spectrometer will not be required.

Derivation of $d \sin \theta = n\lambda$,

where n is the order number.

Applications; e.g. to spectral analysis of light from stars.